VBF-T1LC-BOM-01
Bill of Materials

Case: t1_lc | **Date:** 2026-08-26

Prepared: __________ **Reviewed:** __________ **Approved:** __________

Bill of Materials

Dual caliber: g/cell and kg/kWh. Energy = 17.948 Wh (0.01795 kWh).

Component	Mass (g/cell)	kg/kWh	Source / note
Positive electrode active (NMC811)	16.842	0.938391543326941	mechanical: t x area x (1-eps) x density
Positive conductive additive (carbon black)	0.351	0.01955678848757608	literature default ~2 wt% (estimate)
Positive binder (PVDF)	0.351	0.01955678848757608	literature default ~2 wt% (estimate)
Negative electrode active (graphite)	10.874	0.6058704216920292	mechanical
Negative conductive additive (carbon black)	0.112	0.006240342765266441	literature default ~1 wt% (estimate)
Negative binder (CMC/SBR)	0.224	0.012480685530532882	literature default ~2 wt% (estimate)
Separator (polyolefin)	0.259	0.014430792644678645	mechanical
Electrolyte (EC/EMC + LiPF6)	5.797	0.32299345544865676	pore vol 5.37 cm3 x 1.2 g/cm3 x 0.9 fill (estimate)
Positive current collector (Al)	4.437	0.24721786472756427	mechanical
Negative current collector (Cu)	11.042	0.6152309358399289	mechanical
Enclosure / tabs	Not modeled	Not modeled	beyond parameter set
Total (contract caliber, excl. electrolyte)	43.45	2.4209186888466685	cell/r3_V3_energy.json:mass_kg
Total incl. electrolyte (estimate)	49.25	2.7440792963336804	contract mass + electrolyte estimate